chemistry and chief researcher behind this invention, developed an interest in self-healing materials because of his lifelong love of Wolverine, the comic book character who has the ability to self-heal.

Conventionally, self-healing polymers make use of non-covalent bonds, which create a problem because those bonds are affected by electrochemical reactions that degrade the performance of materials.

Wang helped solve that problem by using a mechanism called ion-dipole interactions, which are forces between charged ions and polar molecules that are highly stable under electrochemical conditions. He combined a polar, stretchable polymer with a mobile, high-ionic-strength salt to create the material with the properties the researchers were seeking.

The material has potential applications in a wide range of fields. It could give robots the ability to self-heal after mechanical failure, extend the lifetime of lithium-ion batteries used in electronics and electric cars, and improve biosensors used in the medical field and environmental monitoring.

## ISRO to share lithium-ion battery technology for electric vehicles

Batteries are the key components of any electric vehicle. At present, all lithium-ion batteries are imported and are quite expensive. Such batteries have high power but weigh less, and their volume is much less as compared to conventional batteries.

Vikram Sarabhai Space Centre under Indian Space Research Organisation (ISRO), India, has developed indigenous technology to manufacture high-power lithium-ion batteries for automobiles and their feasibility tests in vehicles have been successful.

Now, the government has asked ISRO to allow interested players, including from private sector, to obtain their technology for mass production of lithium-ion batteries for electric vehicles. More than half a dozen major automobile companies, battery manufacturers and public sector undertakings have already approached ISRO. These include Mahindra Renault, Hyundai, Nissan, Tata Motors, High Energy Batteries, BHEL and Indian Oil.

## RUUUH by Microsoft is an AI chatbot that talks entertainment

Microsoft has launched an artificially-intelligent chatbot called RUUH. It is targeted at young Indians who are interested in Bollywood, entertainment, music and travel. The bot speaks English and you can interact with it via Facebook Messenger.

Microsoft has been investing in AI and chatbot technologies. A company spokesperson has said, "At Microsoft, we are focused on helping people and organisations achieve more through new conversation models. We are excited about the possibilities in this space and are experimenting with a limited pilot program of a new chatbot that is focused on advancing conversational capabilities within our AI ambition. We hope to expand this chatbot to a broader audience in the future."

RUUH joins the family of chatbots launched by Microsoft, following Tay and Zo. Microsoft had earlier launched Xiaoice, a special chatbot for Chinese audience. It also became very popular and made an appearance on Dragon TV's Morning News.

## Coffee Cookie, a rechargeable device that keeps the coffee warm

*A lightweight disc that attaches to the bottom of disposable coffee cups, the Coffee Cookie is rechargeable for daily use (Image courtesy: http://news.mit.edu)*

Coffee Cookie is a small drink warmer that can heat up to 90°C. It is rechargeable and portable. Massachusetts Institute of Technology (USA) students Gabe Alba and Victoria Gregory came up with this idea, which they believe will bring a major change in people's coffee consumption habits.

The new device is circular and can easily fit in beneath a disposable paper coffee cup. Whether it is coffee or any other drink, the device heats up to maintain its temperature.

Usually, the energy required to keep a drink hot exceeds the battery power capacity for a small portable device, but after conducting an online survey of 300 people, Alba and Gregory found there was a narrower niche to fill. They noted that a number of survey respondents, particularly millennials, tend to abandon takeout coffee when it loses heat, tossing the final third.

In a single day, they designed and 3D-printed, in fast mode, a raw proof of concept. The first prototype lacked a circuit board, and parts of it were held together with hot glue. Next came a series of more sophisticated prototypes that fit a wide variety of takeout cups and included a circuit board, an injection-moulded outer shell, component parts and an official patent-pending status.